Arab academy for Science, Technology and Maritime Transport

Binary Subtraction

Rules of Binary Subtraction

- **0 - 0 = 0**
- **1 - 0 = 1**
- **1 - 1 = 0**
- **0 - 1 = 1 ; Borrow = 1**

When borrowing, 0 becomes 10 (binary), and 10 - 1 = 1.
Subtract from right to left, borrowing when necessary.

Example:

1) $(1011)_2 - (111)_2$

$$\begin{array}{r} {\scriptstyle 0\ 10\ \ \ } \\ 1\ 0\ 1\ 1 \\ -\ \ 1\ 1\ 1 \\ \hline 0\ 1\ 0\ 0 \end{array}$$

2) $(110010)_2 - (101101)_2$

$$\begin{array}{r} {\scriptstyle \ \ 0\ 1\ 10\ 0\ 10} \\ 1\ 1\ 0\ 0\ 1\ 0 \\ -\ 1\ 0\ 1\ 1\ 0\ 1 \\ \hline 0\ 0\ 0\ 1\ 0\ 1 \end{array}$$

3) $(100010)_2 - (10110)_2$

$$\begin{array}{r} {\scriptstyle 0\ 1\ 1\ 10\ \ } \\ 1\ 0\ 0\ 0\ 1\ 0 \\ -\ \ 1\ 0\ 1\ 1\ 0 \\ \hline 0\ 0\ 1\ 1\ 0\ 0 \end{array}$$

Subtract the following binary numbers:

1. $(11101)_2 - (110)_2$
2. $(1000)_2 - (0111)_2$
3. $(1010)_2 - (101)_2$
4. $(1100)_2 - (1011)_2$
5. $(110000)_2 - (11110)_2$
6. $(1010)_2 - (111)_2$
7. $(1100)_2 - (0010)_2$

Subtract the following decimal numbers using 8-bit Two's Complement binary representation:

1. 5 - 3
2. 7 - 5
3. (-2) - 5
4. 13 - 9
5. (-17) - 2
6. (-64) - 32

Reminder: Represent each operand and the result using exactly 8 bits. Use Two's Complement to represent negative numbers and perform subtraction (e.g., 5 = 00000101, -2 = 11111110).